

■ ProductService.java – Full Tanglish Explanation

package com.inventory.service;

➡■ Ithu namma service layer ku related package – logic & database kitta connection illa, pura logic definition maattum tha irukum.

import com.inventory.model.Product;

➡■ Product class-a import pannirukom. Adha use pannum methods la.

import java.util.List; import java.util.Optional;

➡■ List – multiple Product objects return panna use aagum.

➡■ Optional – null pointer exception avoid panna use aagum.

public interface ProductService { ... }

➡■ Interface na blueprint maari. Implementation la actual logic irukum (ex: ProductServiceImpl).

➡■ Service layer abstraction kudukkum – controller direct database-a touch panna kudathu.

List<Product> getAllProducts();

➡■ Ithu ellaa products-a list format la return pannum method.

➡■ Controller la display panna use aagum.

Product saveProduct(Product product);

➡■ Oru pudhu product add panna illa update panna idha use pannuvom.

➡■ Save na repository.save(product) ku mapping.

Optional<Product> getProductById(Long id);

➡■ Specific product one id la thedi eduka use aagum.

➡■ Optional use pannathu safety ku – product illa na null exception varathu.

void deleteProduct(Long id);

➡■ Id kuduthu product delete panna use pannum method.

■ Summary:

■ Ithu oru **Interface layer** – product related CRUD operations define pannum.

■ Real implementation la logic irukum (ProductServiceImpl).

- Structure clear-a maintain panna, readable & reusable code kidaikum.